Module 6

How to Choose Projects

FINE 3010, Financial Management

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What Is Financial Management?

Recall from **Module 1**

- How do businesses achieve shareholder value maximization?
- Small (coffee shop) or large (Tesla), success requires ***sound financial management***, good decisions involving money
- Financial management = **Investment decisions** + **Financing decisions**



How to Maximize Shareholder Value?

Recall from **Module 1**

- Investment Decision
  - Choose investments whose returns are higher than any other alternative use of cash
- Financing Decision
  - Choose the cheapest source of funding

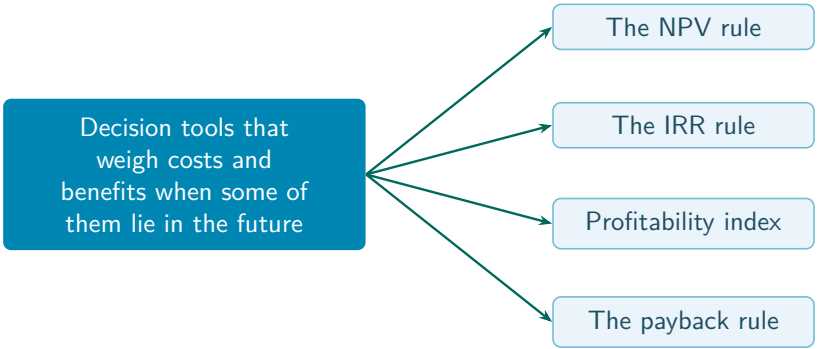


Outline

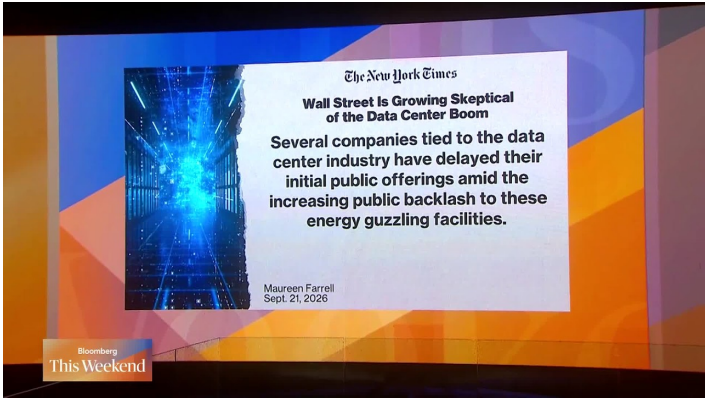
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Outline

- **Capital expenditure**, the amount spent on buying capital (a business's fixed assets), in the hope of generating more cash later
- What capital expenditure should we make?



Motivating Example, Data Centers



Bloomberg, September 27, 2026 (5:55), "Data Center Boom Meets Wall Street Skepticism"

Module 6, Part 1

Net Present Value

Investment Decision-Making Rules

- **Net present value (NPV)**,

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Investment Decision-Making Rules

- **Net present value (NPV)**,

$$\text{NPV} = \text{PV(Benefits)} - \text{PV(Costs)}$$

- "Net" because we evaluate the difference between benefits and costs
- Often (but not necessarily),
  - The main cost is immediate (e.g., build a factory, develop a product), so it needs no discounting
  - Most benefits occur in the future,

$$\text{NPV} = -\text{Initial investment} + \text{PV of future cash flows}$$

The NPV Rule

- **NPV rule**,
  - Take all decisions with NPV > 0
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- **NPV rule**,
  - Take all decisions with NPV > 0
  - Reject all decisions with NPV < 0
- For a business, accepting a project with an NPV of x changes the shareholder value of the business by x

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1. Forecast all current and future (expected) cash flows from the project, both benefits and costs

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$$\text{NPV} = \text{CF}_0 + \frac{\text{CF}_1}{(1+r)^1} + \frac{\text{CF}_2}{(1+r)^2} + \cdots + \frac{\text{CF}_T}{(1+r)^T}$$

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4. Pursue the project if the NPV is positive

Example, Net Present Value

- You are a commercial landlord with the opportunity to buy a new office building
- A tenant is lined up who will generate $25,000 per year in cash flows for three years
- At the end of three years you expect to sell the building for $450,000
- The building costs $375,000, and the opportunity cost is 7%



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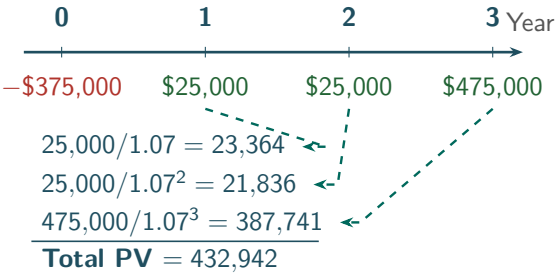


Example, Net Present Value

- Cost ($t = 0$), \$375,000
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- Sale price ($t = 3$), \$450,000
- Opportunity cost, 7%

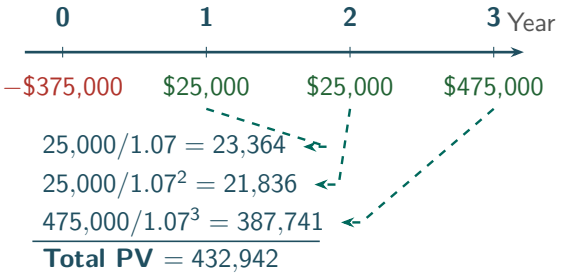
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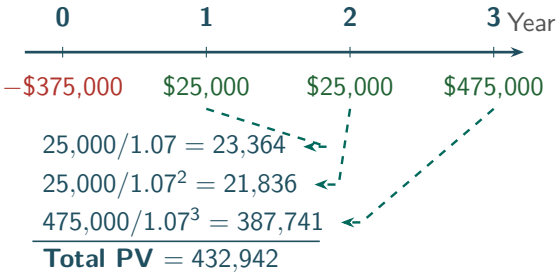
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$$\text{NPV} = -\$375{,}000 + \frac{\$25{,}000}{(1+0.07)^1} + \frac{\$25{,}000}{(1+0.07)^2} + \frac{\$475{,}000}{(1+0.07)^3} = \$57{,}942$$

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- The NPV is positive, so go ahead and purchase the building

Example, NPV with a Financial Calculator (TI BA II Plus)

- Cost ($t = 0$), \$375,000
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Enter	Keys
	CF, **2nd**, **CLR Work**
−375	**ENTER**, ↓
25	**ENTER**, ↓ ↓
25	**ENTER**, ↓ ↓
475	**ENTER**
	NPV (the calculator asks for I)
7	**ENTER**, ↓
	CPT
57.94	

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- Cost ($t = 0$), \$375,000
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- Sale price ($t = 3$), \$450,000
- Opportunity cost, 7%
- The NPV is \$57.94 thousand

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- Solution, now

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- You should not do the project, because the NPV is negative

NPV Profile

- **NPV profile**, the graphical relation between NPV and r

NPV Profile

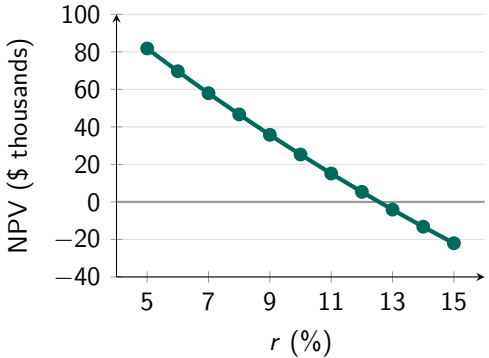
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- **NPV profile**, the graphical relation between NPV and r
- Typically (but not always) downward sloping
- E.g., the building project, cost \$375,000, cash flows \$25,000 in years 1–3, sale price \$450,000. Plot the NPV profile
- Solution, list different values of r, compute the NPV for each, and plot



Back to Data Centers, Each Dollar Must Earn 40 Cents a Year

- Amazon, Alphabet, Microsoft, Meta, and Oracle spent $381 billion on capital expenditure in 2025, about double that is forecast for 2026

Source, Wachter and Wachter, "What Investment Data Implies about the AI Transition," NBER Working Paper 35290, 2026

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- AI investment is risky, investors require about a 15% return (the opportunity cost)
- To break even, each dollar of AI capital must produce about 40 cents of profit a year, 25 to replace the chips plus 15 for the investors
- **"These firms risk bankruptcy unless expected profits grow commensurately"**

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Example, Valuing Non-Constant Growth

Recall from **Module 5**

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	$2	$2	$2	$3	$3

Example, Valuing Non-Constant Growth



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- We discounted the year-5 price of the constant-growth stream, then the five dividends, term by term,

$$\text{PV}_{\text{Price in year 5}} = \frac{3 \times 1.05}{0.10 - 0.05} \times \frac{1}{1.10^5} = \$39.12$$

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$$\text{PV}_{\text{Dividends 1–5}} = \frac{2}{1.10} + \frac{2}{1.10^2} + \frac{2}{1.10^3} + \frac{3}{1.10^4} + \frac{3}{1.10^5} = \$8.89$$

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- Total, $\text{PV}_{\text{Stock}} = 39.12 + 8.89 = \$48.00$

Example, Non-Constant Growth with a Financial Calculator



$$+ \frac{3 \times 1.05}{0.10 - 0.05}$$

Example, Non-Constant Growth with a Financial Calculator



- Enter the cash flows one by one, 0, 2, 2, 2, 3, 66
- In year 5, enter the dividend plus the price, $3 + 63 = 66$

Enter	Keys
	CF, **2nd**, **CLR Work**
0	**ENTER**, ↓
2	**ENTER**, ↓ ↓
2	**ENTER**, ↓ ↓
2	**ENTER**, ↓ ↓
3	**ENTER**, ↓ ↓
66	**ENTER**
	NPV (the calculator asks for I)
10	**ENTER**, ↓
	CPT
48.00	

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3	ENTER, ↓ ↓
66	ENTER
	NPV (the calculator asks for I)
10	ENTER, ↓
	CPT
48.00	

- Enter the cash flows one by one, 0, 2, 2, 2, 3, 66
- In year 5, enter the dividend plus the price, $3 + 63 = 66$
- **No need to discount each cash flow and sum by hand!**

Example, Buying the Stock Is an NPV Decision

- The same stock (dividends 2, 2, 2, 3, 3, then 5% growth, required return on equity 10%) sells for $50 today
- Should you buy it?

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$$\text{NPV} = -50 + \text{PV}(\text{Dividends}) = -50 + 48.00 = -\$2.00$$

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- On the calculator, the same worksheet with -50 entered as the first cash flow (instead of 0)

Practice Problem

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- It will then provide cash flows of $1,250 at the end of each year for 6 years

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- It will then provide cash flows of $1,250 at the end of each year for 6 years
- What is the NPV if the discount rate is 13%?

Practice Problem

- Solution, first draw the timeline



Practice Problem

- Solution 1 of 3, enter the cash flows one at a time

Practice Problem

- Solution 1 of 3, enter the cash flows one at a time

Enter	Keys
	CF, 2nd, CLR Work
−4000	ENTER, ↓
1250	ENTER, ↓ ↓
1250	ENTER, ↓ ↓
1250	ENTER, ↓ ↓
1250	ENTER, ↓ ↓
1250	ENTER, ↓ ↓
1250	ENTER
	NPV (the calculator asks for I)
13	ENTER, ↓
	CPT
996.93	

Practice Problem

- Solution 1 of 3, enter the cash flows one at a time
- The NPV is \$997, take the project

Enter	Keys
	CF, 2nd, CLR Work
−4000	ENTER, ↓
1250	ENTER, ↓ ↓
1250	ENTER, ↓ ↓
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Practice Problem

- Solution 2 of 3, enter the cash flow once with its frequency

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Enter	Keys
	CF, 2nd, CLR Work
−4000	ENTER, ↓
1250	ENTER, ↓
6	ENTER (the frequency F01)
	NPV (the calculator asks for I)
13	ENTER, ↓
	CPT
996.93	

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Enter	Keys
	2nd, CLR TVM
6	N
13	I/Y
1250	PMT
0	FV
	CPT, PV
−4996.93	

Practice Problem

- Solution 3 of 3, the cash flows are an annuity, use the TVM keys
- Change the sign to positive, subtract the 4,000 cost, the NPV is $997, take the project

Enter	Keys
	2nd, CLR TVM
6	N
13	I/Y
1250	PMT
0	FV
	CPT, PV
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Recap of Part 1

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Recap of Part 1

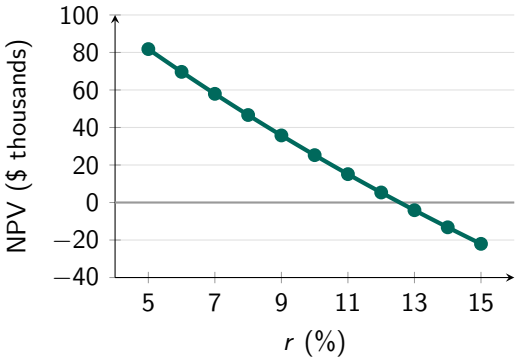
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- **NPV profile**, the graphical relation between NPV and r

Module 6, Part 2

Internal Rate of Return

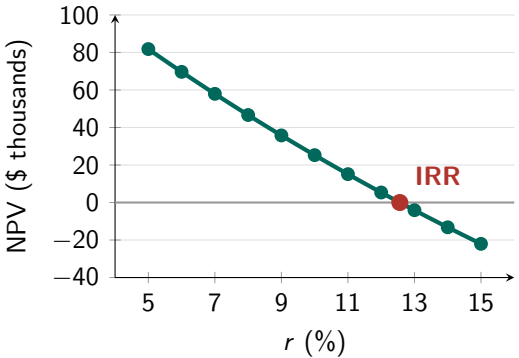
Internal Rate of Return

- **Internal rate of return (IRR)**, the discount rate r at which $\text{NPV} = 0$



Internal Rate of Return

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- On the NPV profile, the IRR is where the curve crosses zero, here around 12.5%



Internal Rate of Return

- Find the IRR by solving the equation

$$\text{NPV} = 0 = \text{CF}_0 + \frac{\text{CF}_1}{(1+\text{IRR})^1} + \frac{\text{CF}_2}{(1+\text{IRR})^2} + \cdots + \frac{\text{CF}_T}{(1+\text{IRR})^T}$$

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- E.g., the building project, cost \$375,000, cash flows \$25,000 in years 1–3, sale price \$450,000. What is the IRR?
- Solution,

$$0 = -\$375{,}000 + \frac{\$25{,}000}{(1+\text{IRR})^1} + \frac{\$25{,}000}{(1+\text{IRR})^2} + \frac{\$475{,}000}{(1+\text{IRR})^3}$$

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25	**ENTER**, ↓ ↓
25	**ENTER**, ↓ ↓
475	**ENTER**
	IRR (shows 0)
	CPT
12.56%	

Example, IRR with a Financial Calculator (TI BA II Plus)

- Cost ($t = 0$), \$375,000
- Cash flows in years 1–3, \$25,000
- Sale price ($t = 3$), \$450,000
- The IRR is 12.56%

Enter	Keys
	CF, **2nd**, **CLR Work**
−375	**ENTER**, ↓
25	**ENTER**, ↓ ↓
25	**ENTER**, ↓ ↓
475	**ENTER**
	IRR (shows 0)
	CPT
12.56%	

Internal Rate of Return, a Pitfall

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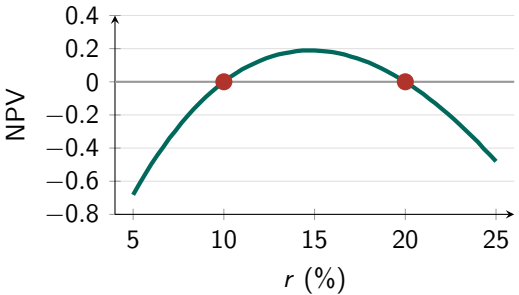


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- Two IRRs, at 10% and 20%



The IRR Rule

- **IRR rule**,
  - If IRR $>$ opportunity cost r, accept the project
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- The IRR rule often gives the same answer as the NPV rule
- With multiple IRRs, use the NPV rule instead of the IRR rule

Example, the IRR Rule

- Apply the IRR rule to the building project if,
  - A. the opportunity cost is 7%
  - B. the opportunity cost is 14%

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- Solution,
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- The NPV rule gives the same answers here, NPV > 0 in A and NPV < 0 in B

Practice Problem

- The same project as before, $4,000 to set up, then cash flows of $1,250 at the end of each year for 6 years

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- Solution, the same cash flow worksheet, then **IRR**, **CPT**

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Enter	Keys
	CF, **2nd**, **CLR Work**
−4000	**ENTER**, ↓
1250	**ENTER**, ↓
6	**ENTER** (the frequency F01)
	IRR (shows 0)
	CPT
21.57%	

Practice Problem

- Solution, the same cash flow worksheet, then **IRR**, **CPT**
- The IRR is 21.6%, reject once the discount rate exceeds it

Enter	Keys
	CF, **2nd**, **CLR Work**
−4000	**ENTER**, ↓
1250	**ENTER**, ↓
6	**ENTER** (the frequency F01)
	IRR (shows 0)
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Practice Problem

- A business believes its latest project, which will cost $99,000 to set up, will generate a perpetual growing stream of cash flows
- Cash flow at the end of the first year will be $7,000, and cash flows in future years are expected to grow indefinitely at an annual rate of 5%

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a. If the discount rate for this project is 10%, what is the project NPV?
b. What is the project IRR?

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$$\text{NPV} = -99{,}000 + \frac{7{,}000}{0.10 - 0.05} = -99{,}000 + 140{,}000 = \$41{,}000$$

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$$-99{,}000 + \frac{7{,}000}{\text{IRR} - 0.05} = 0 \qquad \text{so} \qquad \text{IRR} = 0.05 + \frac{7{,}000}{99{,}000} = 12.07\%$$

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- IRR (12.07%) $>$ discount rate (10%), accept the project, the same answer as the NPV rule

Recap of Part 2

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- **Internal rate of return (IRR)**, the discount rate at which NPV $= 0$, where the NPV profile crosses zero
- **IRR rule**, accept if IRR $> r$, reject if IRR $< r$
- Cash flows that change sign more than once can have multiple IRRs, when the rules disagree, trust the NPV rule

Module 6, Part 3

Choosing among Projects

Mutually Exclusive Projects

- What if you must choose between mutually exclusive projects (you can only choose **one**)? Then,
  1. Calculate the NPV of each project
  2. Among the options with a positive NPV, choose the one with the highest NPV

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  - The IRR ignores the size of a project
  - NPV captures "dollars", IRR captures "percentage returns", and with mutually exclusive projects you care about maximizing dollars

Example, Mutually Exclusive Projects

- You choose between Project A and Project B, only one. Both have an opportunity cost of 10%

Cash flows	CF_0	CF_1	CF_2	CF_3
Project A	−100,000	60,000	60,000	60,000
Project B	−500,000	250,000	250,000	250,000

a. Calculate the NPV and IRR of each project

b. Under the NPV rule and the IRR rule, are these good projects?

c. Which project should you pick if you can only pick one?

Example, Mutually Exclusive Projects

- Solution,

	NPV	IRR
Project A	$49,211	36%
Project B	$121,713	23%

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Project A	\$49,211	36%
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- b. Both projects are good under both rules (NPV > 0 and IRR $> r$ for both)
- c. Pick Project B, its NPV is higher (even though the IRR is higher for Project A)

Profitability Index

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  - Go down the ranking
  - Stop when you run out of dollars or run out of positive-NPV projects

Example, Profitability Index

- You choose among Projects A, B, and C, all with an opportunity cost of 10%
- But you only have $10,000 in total to invest

Cash flows	CF_0	CF_1	CF_2
Project A	−10,000	30,000	5,000
Project B	−5,000	5,000	20,000
Project C	−5,000	5,000	15,000

a. Calculate the NPV, IRR, and profitability index of each project

b. Under the NPV rule and the IRR rule, are these good projects?

c. Which project(s) should you pick if you can only invest $10,000?

Example, Profitability Index

- Solution,

	NPV	IRR	PI
Project A	$21,405	216%	2.1
Project B	$16,074	156%	3.2
Project C	$11,942	130%	2.4

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- c. With only $10,000, rank on the profitability index and go down the list,
  - First pick Project B (highest PI), $5,000 left

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- **Capital rationing**, a limit on the amount of funds available for investment
- **Soft rationing**, limits on available funds imposed by management
- **Hard rationing**, limits on available funds imposed by the unavailability of funds in the capital market
  - **Capital market**, the financial markets where businesses raise long-term funds, such as the stock and bond markets

Recap of Part 3

- Among mutually exclusive projects, take the positive-NPV project with the highest NPV

Recap of Part 3

- Among mutually exclusive projects, take the positive-NPV project with the highest NPV
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Recap of Part 3

- Among mutually exclusive projects, take the positive-NPV project with the highest NPV
- **Profitability index**, Net present value / Initial investment
- Under capital rationing, rank projects by their profitability index and go down the ranking

Module 6, Part 4

Payback Period and Payback Rule

Payback Period

- **Payback period**, the number of years until the cash flows recover the initial investment of the project

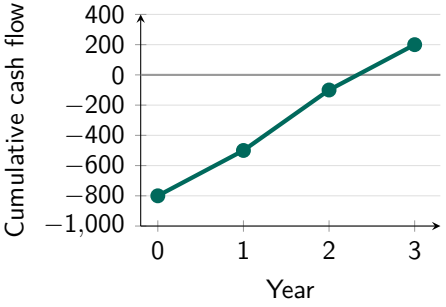
Payback Period

- **Payback period**, the number of years until the cash flows recover the initial investment of the project
- E.g., consider Project X, what is its payback period?

Cash flows	$t = 0$	$t = 1$	$t = 2$	$t = 3$
Project X	-800	300	400	300

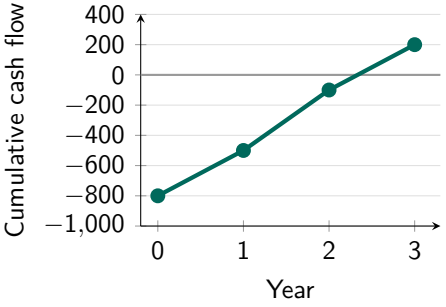
Example, Payback Period

- Solution, find the time until the cumulative cash flows (CCF) turn positive,
  - After $t = 0$, CCF $= -800$
  - After $t = 1$, CCF $= -500$
  - After $t = 2$, CCF $= -100$
  - After $t = 3$, CCF $= +200$



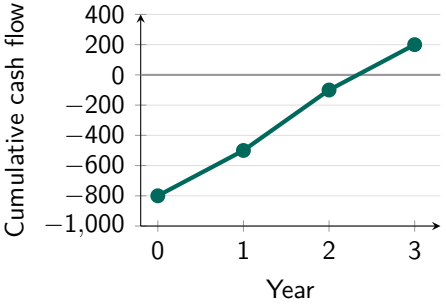
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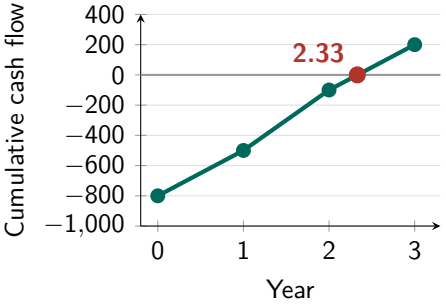
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- The payback period is 2.33 years



Payback Rule

- **Payback rule**, first define a maximum allowed payback period as a cutoff,
  - If payback period $<$ maximum cutoff, accept the project
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- E.g., Project X again. The company's cutoff is 2 years. Accept or reject under the payback rule?

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Cash flows	CF_0	CF_1	CF_2	CF_3
Project X	-800	300	400	300

- Solution, payback period (2.33) $>$ maximum cutoff (2), reject the project
- What if the maximum cutoff had been 3 years instead? Then accept

Payback Rule, Pros and Cons

Pros 👍

- Easy to understand and compute

Cons 👎

- Does not consider the time value of money, as there is no discounting
- Only considers cash flows up to the cutoff date, what if really large cash flows come after it?
- The maximum cutoff is arbitrary

Practice Problem

- When calculating a project's payback period, cash flows are,

A. discounted at the opportunity cost
B. discounted at the internal rate of return
C. discounted at the risk-free rate of return
D. not discounted at all

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- A project's payback period is determined to be 4 years. If it is later discovered that additional cash flows will be generated in years 5 and 6, then the project's payback period will,

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D. change, but the discount rate must be known to determine the nature of the change

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Payback Rule Is Often Not Good at Ranking Projects

- Consider Projects A and B. The company's payback cutoff is 2 years, and the opportunity cost for all projects is 10%

Cash flows	$t = 0$	$t = 1$	$t = 2$	$t = 3$
Project A	−2,000	500	1,000	3,000
Project B	−2,000	2,000	0	0

a. Calculate the payback period, NPV, and IRR of each project

b. Under the payback rule, which project(s) should the company take?

c. Why is the payback rule's choice wrong?

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- The payback period is $2 + 0.17 = 2.17$ years

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- Project B,
  - After $t = 0$, CCF $= -2{,}000$
  - After $t = 1$, CCF $= 0$
  - After $t = 2$, CCF $= 0$
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  - After $t = 1$, CCF $= 0$
  - After $t = 2$, CCF $= 0$
  - After $t = 3$, CCF $= 0$

- The cash flows recover the initial investment right at $t = 1$, so the payback period is 1.00 year

Payback Rule Is Often Not Good at Ranking Projects

- Solution,

	Payback period	NPV	IRR
Project A	2.17	1,535	39%
Project B	1.00	−182	0%

Payback Rule Is Often Not Good at Ranking Projects

- Solution,

	Payback period	NPV	IRR
Project A	2.17	1,535	39%
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- b. Only Project B has a payback period (1.00) below the cutoff (2 years), so the payback rule takes only Project B

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- Solution,

	Payback period	NPV	IRR
Project A	2.17	1,535	39%
Project B	1.00	−182	0%

- b. Only Project B has a payback period (1.00) below the cutoff (2 years), so the payback rule takes only Project B
- c. The payback rule is wrong, on NPV Project A is the best choice, while Project B should be rejected because its NPV is negative

Payback Rule Is Often Not Good at Ranking Projects

- Solution,

	Payback period	NPV	IRR
Project A	2.17	1,535	39%
Project B	1.00	−182	0%

- b. Only Project B has a payback period (1.00) below the cutoff (2 years), so the payback rule takes only Project B
- c. The payback rule is wrong, on NPV Project A is the best choice, while Project B should be rejected because its NPV is negative

Large cash flows that come after the cutoff are ignored

Recap of Part 4

- **Payback period**, the number of years until the cash flows recover the initial investment

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- **Payback period**, the number of years until the cash flows recover the initial investment
- **Payback rule**, accept if the payback period is shorter than a maximum cutoff
- Easy to compute, but it ignores the time value of money and every cash flow after the cutoff, so it can rank projects wrongly

Module 6, Part 5

Decision Rules in Practice

Capital Budgeting Techniques, Summary

Criterion	Definition	Investment rule
Net present value (NPV)	Present value of benefits minus present value of costs	Take all projects with NPV > 0, reject all with NPV < 0
Internal rate of return (IRR)	The discount rate at which NPV $= 0$	Accept if IRR $>$ opportunity cost r, reject if IRR $< r$
Profitability index	Net present value divided by initial investment	Under capital rationing, rank projects by their profitability index and go down the ranking
Payback period	The number of years until the cash flows recover the initial investment	Accept if the payback period is shorter than a maximum cutoff

Facing Multiple Projects, Summary

Situation	What to do
No constraint	Take all projects with NPV $>$ 0, reject all with NPV $<$ 0
Can only choose one among mutually exclusive projects	Take the positive-NPV project with the highest NPV, not the highest IRR
Limited amount of funds for the initial investment (capital rationing)	Rank projects by their profitability index from highest to lowest and go down the ranking, stop when you run out of dollars or run out of positive-NPV projects

Capital Budgeting Techniques, Use in Practice

Investment criterion	% of firms that always or almost always use it
Internal rate of return	76
Net present value	75
Payback period	57
Profitability index	12

Source, J. R. Graham and C. R. Harvey, "The Theory and Practice of Corporate Finance, Evidence from the Field," *Journal of Financial Economics*, 2001

Practice Problem

- When managers select correctly from among mutually exclusive projects, they,

A. may give up rate of return for NPV
B. may give up NPV for rate of return
C. select the largest project
D. focus on the payback method to avoid conflicting signals

Practice Problem

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Practice Problem

- Which of the following investment criteria takes the time value of money into consideration?

A. Net present value only
B. Profitability index and net present value only
C. Internal rate of return and net present value only
D. Profitability index, internal rate of return, and net present value

Practice Problem

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A. Net present value only
B. Profitability index and net present value only
C. Internal rate of return and net present value only
D. **Profitability index, internal rate of return, and net present value**

Practice Problem

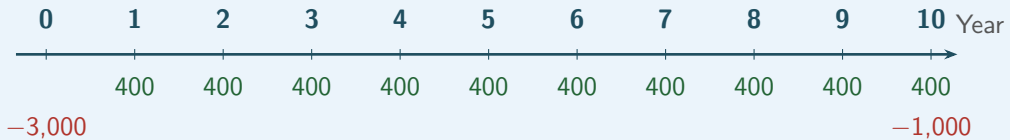
- Building a nuclear power plant costs $3 billion now
- The plant will then provide cash flows of $400 million at the end of each year for 10 years
- At the end of year 10, the plant must also be shut down at a cost of $1 billion

Practice Problem

- Building a nuclear power plant costs $3 billion now
- The plant will then provide cash flows of $400 million at the end of each year for 10 years
- At the end of year 10, the plant must also be shut down at a cost of $1 billion
- What is the NPV of the project if the discount rate is 5%?

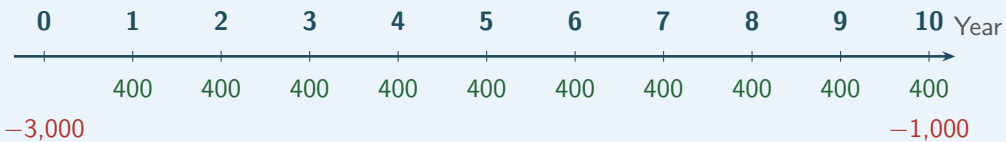
Practice Problem

- Solution, first draw the timeline (in $ millions)



Practice Problem

- Solution, first draw the timeline (in $ millions)



- In the final year, the cash flow is $400 - 1{,}000 = -\$600$ million

Practice Problem

- Then the cash flow worksheet (in $ millions)

Practice Problem

- Then the cash flow worksheet (in $ millions)
- Enter 400 once, with frequency 9, then the final −600

Enter	Keys
	CF, **2nd**, **CLR Work**
−3000	**ENTER**, ↓
400	**ENTER**, ↓
9	**ENTER**, ↓ (the frequency F01)
−600	**ENTER**
	NPV (the calculator asks for I)
5	**ENTER**, ↓
	CPT
−525.22	

Practice Problem

- Then the cash flow worksheet (in $ millions)
- Enter 400 once, with frequency 9, then the final −600
- The NPV is −$525 million, reject the project

Enter	Keys
	CF, **2nd**, **CLR Work**
−3000	**ENTER**, ↓
400	**ENTER**, ↓
9	**ENTER**, ↓ (the frequency F01)
−600	**ENTER**
	NPV (the calculator asks for I)
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- The NPV rule is the gold standard, take every project with NPV > 0, and among mutually exclusive projects take the highest NPV
- The IRR rule usually agrees with NPV, but it ignores project size and can break down with multiple IRRs
- Under capital rationing, rank projects by the profitability index
- The payback rule is quick but ignores discounting and later cash flows

Thank you!

Please let me know if you have any questions.